

Problem set 1

1. In $\mathbb{R}^n$, show that

(a) $2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2$.

(b) $\|x - y\|\|x + y\| \leq \|x\|^2 + \|y\|^2$.

(c) $4(x \cdot y) = \|x + y\|^2 - \|x - y\|^2$.

$$(a) \quad 2\|x\|^2 + 2\|y\|^2$$

$$= 2(x \cdot x) + 2(y \cdot y)$$

$$\|x + y\|^2 + \|x - y\|^2$$

$$= (x + y) \cdot (x + y) + (x - y) \cdot (x - y)$$

$$= x \cdot x + 2x \cdot y + y \cdot y + x \cdot x - 2x \cdot y + y \cdot y$$

$$= 2(x \cdot x) + 2(y \cdot y)$$

$$\therefore 2\|x\|^2 + 2\|y\|^2 = \|x + y\|^2 + \|x - y\|^2$$

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(c) $4(x \cdot y) = \|x + y\|^2 - \|x - y\|^2.$

$$(b) -2\|x - y\|\|x + y\|$$

$$= (\|x - y\| - \|x + y\|)^2 - (\|x - y\|^2 + \|x + y\|^2)$$

$$= (\|x - y\| - \|x + y\|)^2 - (2\|x\|^2 + 2\|y\|^2)$$

$$\therefore -2\|x - y\|\|x + y\| + 2(\|x\|^2 + \|y\|^2)$$

$$= (\|x - y\| + \|x + y\|)^2$$

$$\therefore -2\|x - y\|\|x + y\| + 2(\|x\|^2 + \|y\|^2) \geq 0$$

$$\therefore \|x - y\|\|x + y\| \leq \|x\|^2 + \|y\|^2$$

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(c) $4(x \cdot y) = \|x + y\|^2 - \|x - y\|^2.$

$$(c) \quad \|x + y\|^2 - \|x - y\|^2$$

$$= (x + y) \cdot (x + y) - (x - y) \cdot (x - y)$$

$$= x \cdot x + 2x \cdot y + y \cdot y - (x \cdot x - 2x \cdot y + y \cdot y)$$

$$= 4x \cdot y$$

$$= 4(x \cdot y)$$

2. Verify the Cauchy-Schwarz inequality and the triangle inequality for the vectors given below:

(a) $\mathbf{x} = (2, 0, -1)$ and $\mathbf{y} = (4, 0, -2)$

(b) $\mathbf{x} = (1, 0, 2, 6)$ and $\mathbf{y} = (3, 8, 4, 1)$

(c) $\mathbf{x} = (1, -1, 1, -1, 1)$ and $\mathbf{y} = (3, 0, 0, 2, 0)$

Cauchy - Schwarz inequality

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Triangle inequality

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$$

$$(a) |\mathbf{x} \cdot \mathbf{y}| = |(2, 0, -1) \cdot (4, 0, -2)|$$

$$= |2 \cdot 4 + 0 \cdot 0 + (-1) \cdot (-2)|$$

$$= |8 + 0 + 2| = 10$$

$$\|\mathbf{x}\| \|\mathbf{y}\| = \sqrt{2^2 + 0^2 + 1^2} \cdot \sqrt{4^2 + 0^2 + (-2)^2}$$

$$= \sqrt{5} \cdot 2\sqrt{5} = 10$$

$$\therefore |\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$$

2. Verify the Cauchy-Schwarz inequality and the triangle inequality for the vectors given below:

(a) $x = (2, 0, -1)$ and $y = (4, 0, -2)$

(b) $x = (1, 0, 2, 6)$ and $y = (3, 8, 4, 1)$

(c) $x = (1, -1, 1, -1, 1)$ and $y = (3, 0, 0, 2, 0)$

$$\begin{aligned}\|x+y\| &= \sqrt{(2+4)^2 + (0+0)^2 + (-1-2)^2} \\ &= \sqrt{6^2 + 0^2 + (-3)^2} = 3\sqrt{5}\end{aligned}$$

$$\|x\| + \|y\| = \sqrt{5} + 2\sqrt{5} = 3\sqrt{5}$$

$$\therefore \|x+y\| = \|x\| + \|y\|$$

(b) $\|x\|\|y\| = \sqrt{41} \cdot 3\sqrt{10} = 3\sqrt{410}$

$$|x \cdot y| = \sqrt{17}$$

$$\therefore |x \cdot y| < \|x\|\|y\|$$

$$\|x\| + \|y\| = \sqrt{41} + 3\sqrt{10}$$

$$\|x+y\| = \|(4, 8, 6, 7)\| = \sqrt{165}$$

$$\therefore \|x+y\| < \|x\| + \|y\|$$

2. Verify the Cauchy-Schwarz inequality and the triangle inequality for the vectors given below:

(a) $\mathbf{x} = (2, 0, -1)$ and $\mathbf{y} = (4, 0, -2)$

(b) $\mathbf{x} = (1, 0, 2, 6)$ and $\mathbf{y} = (3, 8, 4, 1)$

(c) $\mathbf{x} = (1, -1, 1, -1, 1)$ and $\mathbf{y} = (3, 0, 0, 2, 0)$

$$(c) \quad \|\mathbf{x}\| \|\mathbf{y}\| = \sqrt{5} \cdot \sqrt{13} = \sqrt{65}$$

$$|\mathbf{x} \cdot \mathbf{y}| = 1$$

$$\therefore |\mathbf{x} \cdot \mathbf{y}| < \|\mathbf{x}\| \|\mathbf{y}\|$$

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\| &= \|(4, -1, 1, 1, 1)\| \\ &= 2\sqrt{5} \end{aligned}$$

$$\|\mathbf{x}\| + \|\mathbf{y}\| = \sqrt{5} + \sqrt{13}$$

$$\therefore \|\mathbf{x} + \mathbf{y}\| < \|\mathbf{x}\| + \|\mathbf{y}\|$$

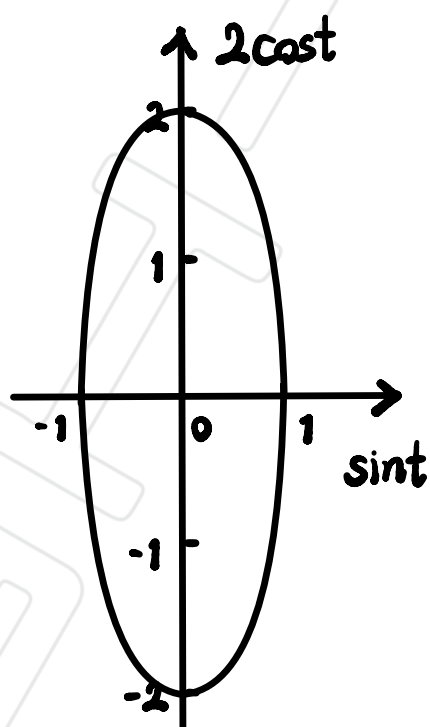
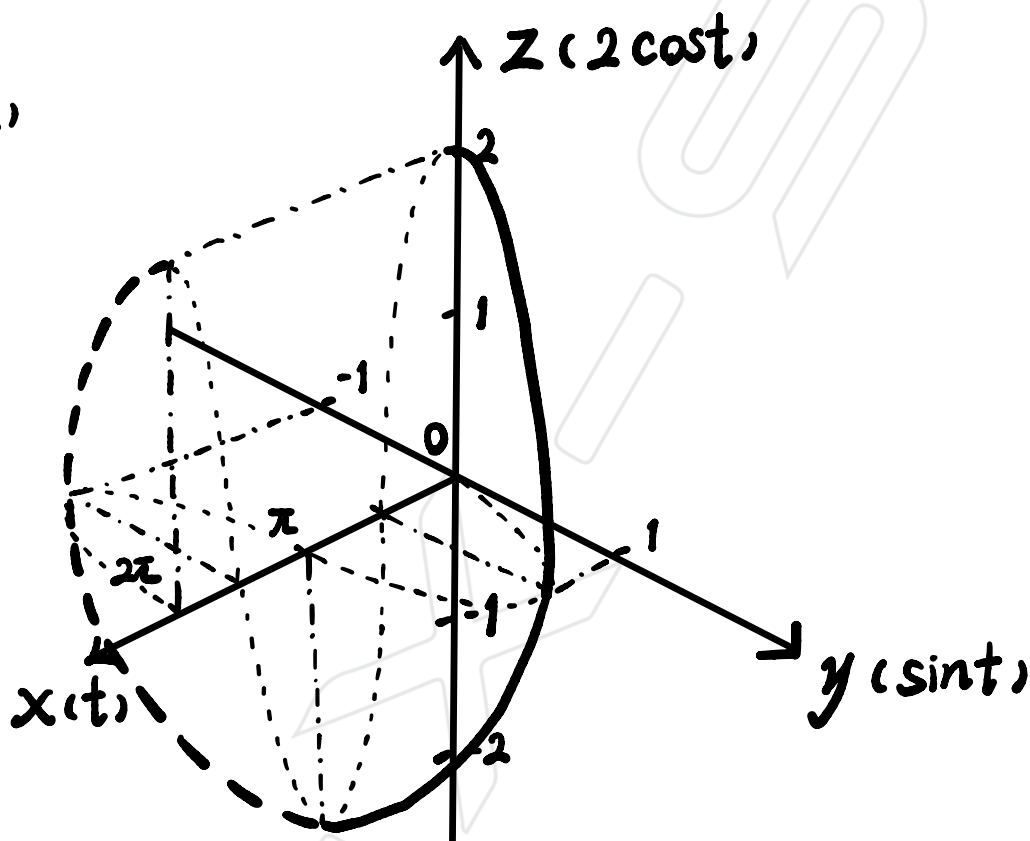
3. Sketch the curves given below and find their tangent vectors at all the points in the curve.

(a) $c(t) = (\sin t, 2 \cos t)$ for $t \in [0, 2\pi]$.

(b) $c(t) = (2t - 1, t + 2, t)$ for $t \in [-2, 2]$.

(c) $c(t) = ((\pi + |t|) \cos t, (\pi + |t|) \sin t)$ for $t \in [-\pi, \pi]$.

(a)



$$\frac{dc}{dt} = (\cos t, -2 \sin t)$$

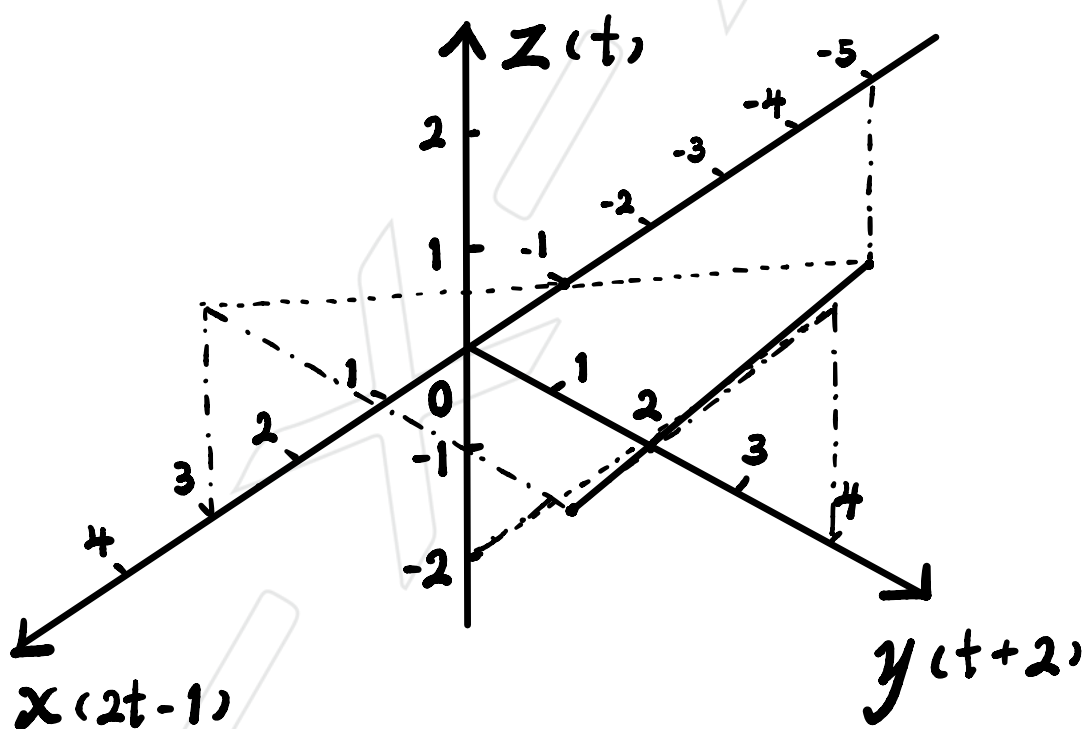
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(b)



$$\frac{d\mathbf{c}}{dt} = (2, 1, 1)$$

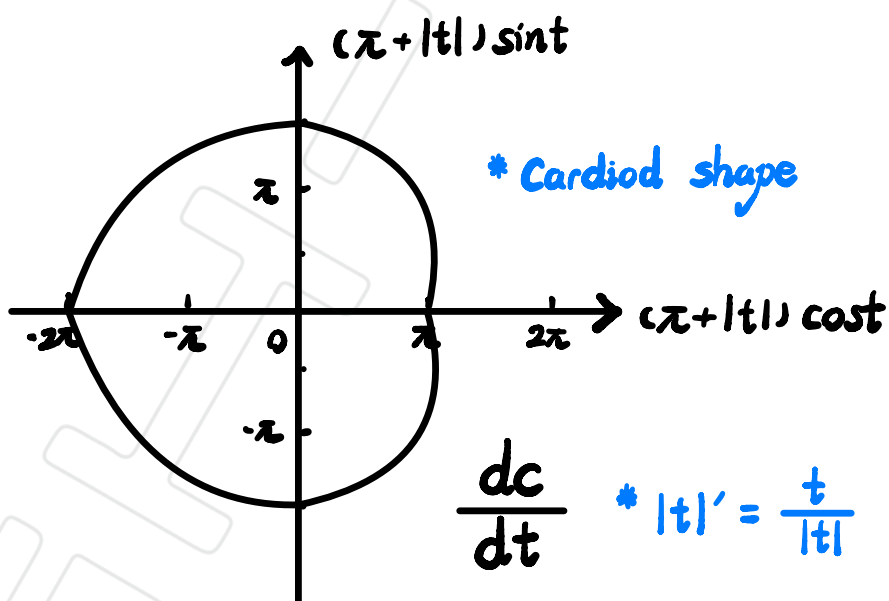
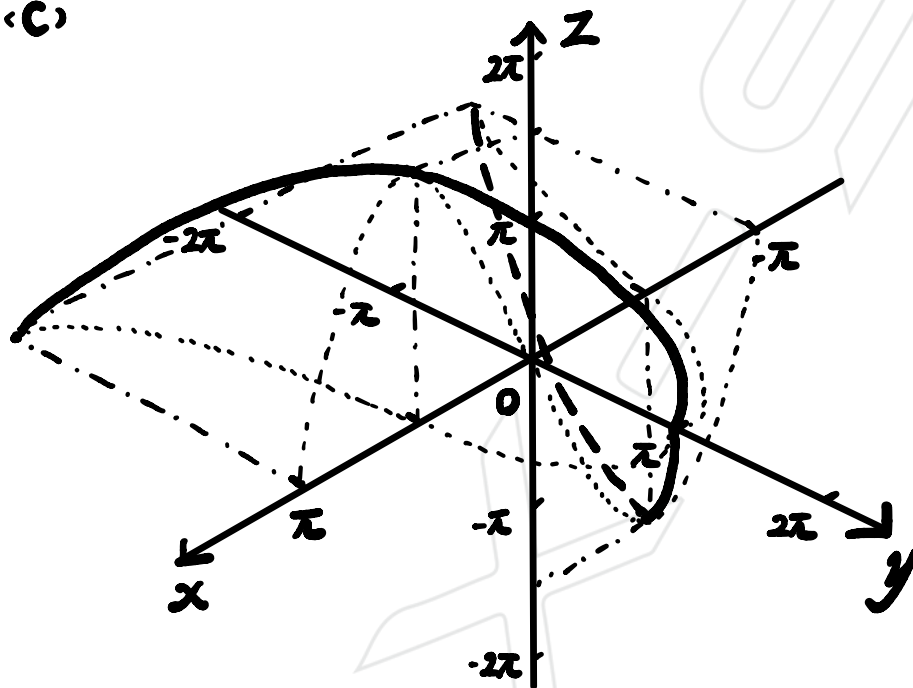
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(c) $\mathbf{c}(t) = ((\pi + |t|) \cos t, (\pi + |t|) \sin t)$ for $t \in [-\pi, \pi]$.

(c)



* Cardioid shape

$$\frac{d\mathbf{c}}{dt} \quad * |t|' = \frac{t}{|t|}$$

$$= (-(\pi + |t|) \sin t, (\pi + |t|) \cos t)$$

$$= (-(\pi + |t|) \sin t + \frac{t}{|t|} \cos t, (\pi + |t|) \cos t + \frac{t}{|t|} \sin t)$$

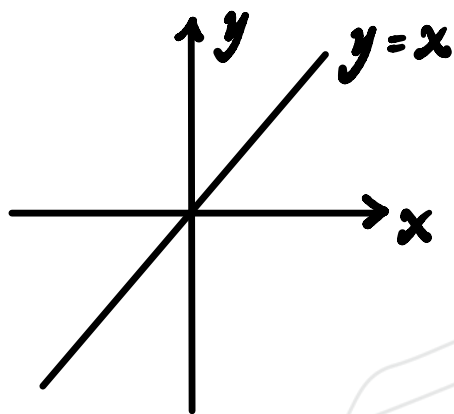
4. Sketch the level sets:

(a) $x - y + z + 2 = c$, where $c = 0, 1, 2, 3$

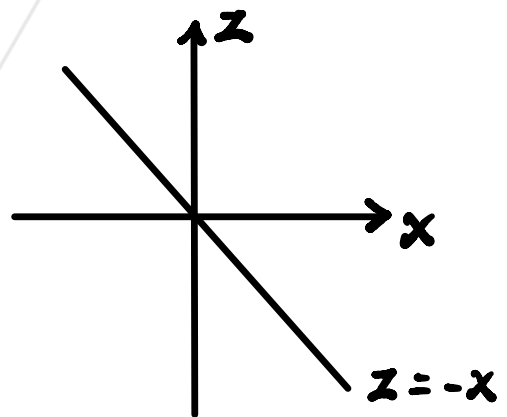
(b) $x^2 - y^2 = c$, where $c = -4, -1, 0, 1, 4$

(c) $x^2 + y^2 + \frac{z^2}{4} = c$, where $c = 0, 1, 4, 9$

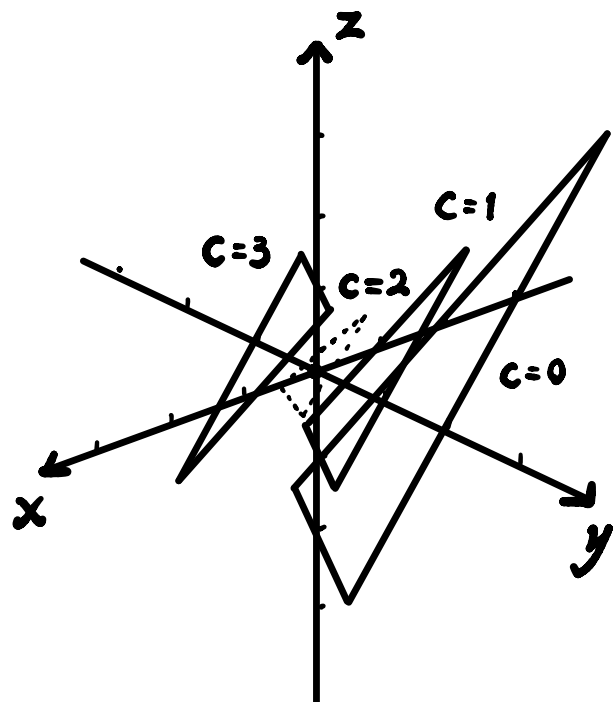
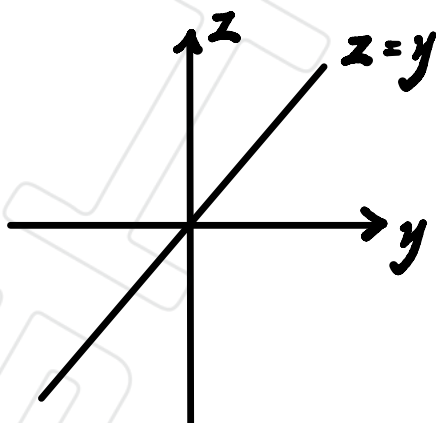
(a) $z = -2, -1, 0, 1$
for $c = 0, 1, 2, 3$



$y = 2, 1, 0, -1$
for $c = 0, 1, 2, 3$



$x = -2, -1, 0, 1$
for $c = 0, 1, 2, 3$



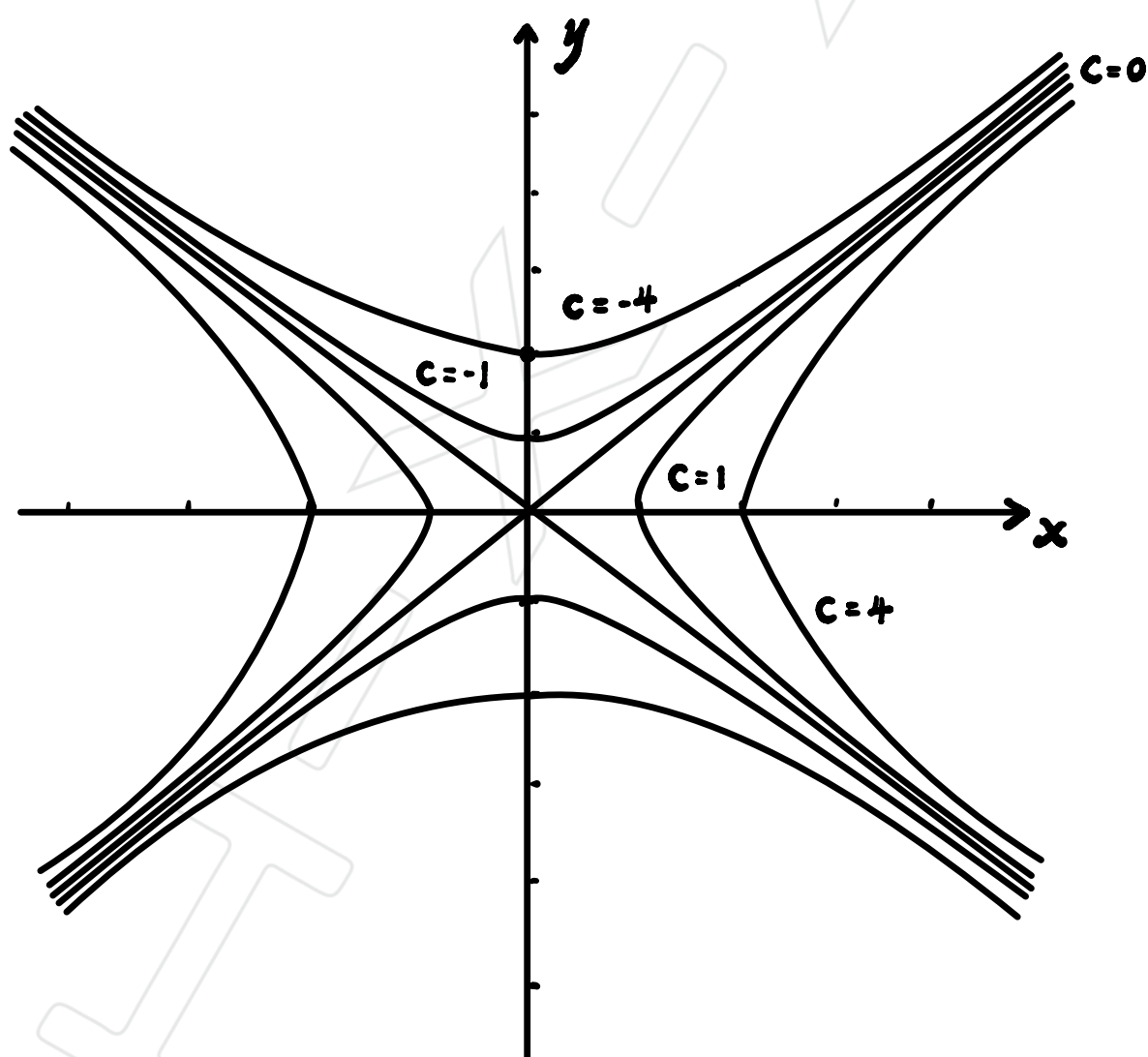
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(c) $x^2 + y^2 + \frac{z^2}{4} = c$, where $c = 0, 1, 4, 9$

(b) $y^2 = x^2 - c$



4. Sketch the level sets:

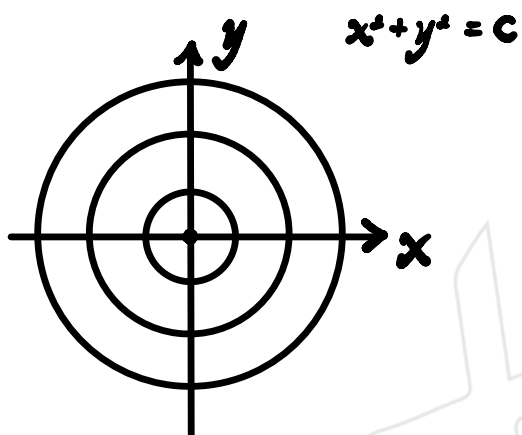
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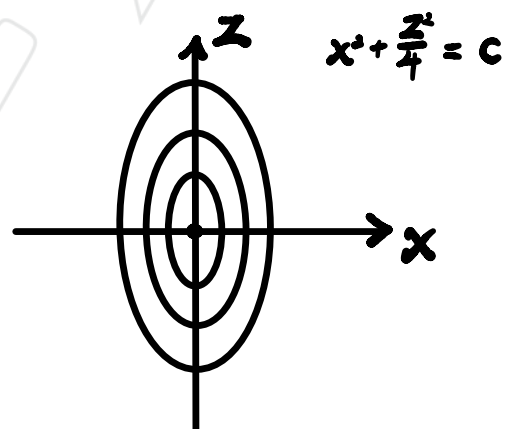
(c) $z = 0$

for $c = 0, 1, 4, 9$



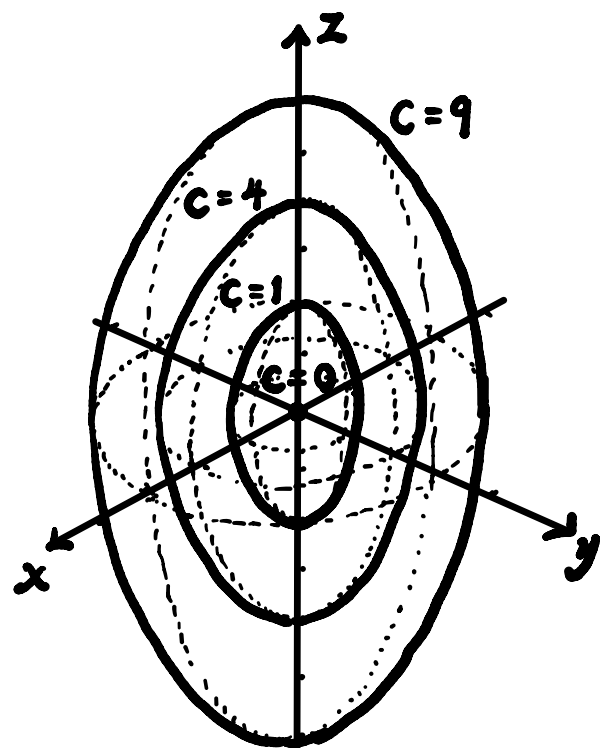
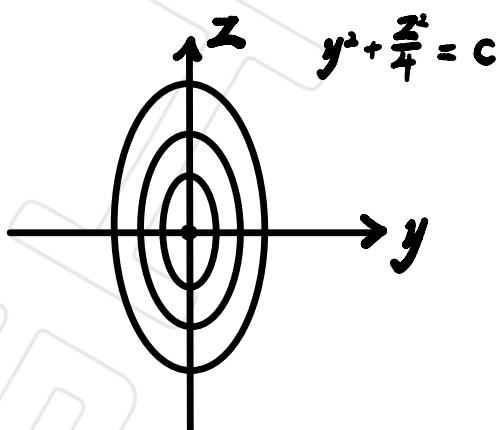
$y = 0$

for $c = 0, 1, 4, 9$



$x = 0$

for $c = 0, 1, 4, 9$



5. Consider a plane S in $\mathbb{R}^3$ given by $ax + by + cz + d = 0$. Show that a unit normal vector to S is given by

$$\mathbf{n} = \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$$

$$\mathbf{n} = \frac{\nabla S}{\|\nabla S\|}$$

$$= \frac{(a, b, c)}{\|(a, b, c)\|}$$

$$= \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$$

6. Find an equation for the tangent plane of $z = f(x, y)$ at the point $(x_0, y_0, f(x_0, y_0))$

(a) $f(x, y) = x - y + 2, (x_0, y_0) = (1, 1),$

(b) $f(x, y) = x^2 + 4y^2, (x_0, y_0) = (2, -1)$

(c) $f(x, y) = xy, (x_0, y_0) = (-1, -1)$

(d) $f(x, y) = x^2 + y^2, (x_0, y_0) = (-1, -1)$

(a) At $(x_0, y_0) = (1, 1), z_0 = 2$

$$\begin{aligned} z &= z_0 + \left. \frac{df}{dx} \right|_{(x_0, y_0)} (x - x_0) + \left. \frac{df}{dy} \right|_{(x_0, y_0)} (y - y_0) \\ &= 2 + (x - 1) + [-(y - 1)] = x - y + 2 \end{aligned}$$

(b) At $(x_0, y_0) = (2, -1), z_0 = 8$

$$\begin{aligned} z &= 8 + (2 \times 2)(x - 2) + (8 \times (-1))(y + 1) \\ &= 4x - 8y - 8 \end{aligned}$$

(c) At $(x_0, y_0) = (-1, -1), z_0 = 1$

$$z = 1 + (-1)(x + 1) + (-1)(y + 1) = -y - x - 1$$

(d) At $(x_0, y_0) = (-1, -1), z_0 = 2$

$$\begin{aligned} z &= 2 + (2 \times (-1))(x + 1) + (2 \times (-1))(y + 1) \\ &= -2x - 2y - 2 \end{aligned}$$

7. The height h of the Hawaiian volcano Mauna Loa is (roughly) described by the function $h(x, y) = 2.59 - 0.00024x^2 - 0.00065y^2$, where h is the height above sea level and x and y measure east-west and north-south distances in miles from the top of the mountain. At $(x, y) = (-2, -4)$:

- (a) How fast is the height increasing in the direction $(1, 1)$? Express your answer in miles of height per a mile of horizontal distance traveled.
- (b) In what direction is the steepest upward path?

$$(a) \frac{df(x_0 + sb)}{ds}$$

$$= \nabla f(x_0) \cdot b$$

$$= (-0.00048x, -0.0013y)_{(x_0, y_0)} \cdot (1, 1)$$

$$= (0.00096, 0.0052) \cdot (1, 1)$$

$$= 0.00096 + 0.0052 = 0.00616$$

$$(b) n = \frac{\nabla f(x_0)}{\|\nabla f(x_0)\|}$$

$$= \frac{(0.00096, 0.0052)}{\sqrt{(0.00096)^2 + (0.0052)^2}}$$

$$= (0.18155, 0.98338)$$

8. Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ if $\mathbf{F}$ is given below:

(a) $\mathbf{F} = 2x\mathbf{i} + 2y\mathbf{j} + 4z\mathbf{k}$

(b) $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$

(c) $\mathbf{F} = (x + y)\mathbf{i} + (y + z)\mathbf{j} + (z + x)\mathbf{k}$

(d) $\mathbf{F} = x\mathbf{i} + 3xy\mathbf{j} + z\mathbf{k}$

$$\begin{aligned} \text{(a)} \quad \nabla \cdot \mathbf{F} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (F_1, F_2, F_3) \\ &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 2 + 2 + 4 = 8 \end{aligned}$$

$$\begin{aligned} \nabla \times \mathbf{F} &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \times (F_1, F_2, F_3) \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_2}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \\ &= (0 - 0, 0 - 0, 0 - 0) = \mathbf{0} \end{aligned}$$

$$\text{(b)} \quad \nabla \cdot \mathbf{F} = 2x + 2y + 2z$$

$$\nabla \times \mathbf{F} = (0 - 0, 0 - 0, 0 - 0) = \mathbf{0}$$

$$\text{(c)} \quad \nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$$

$$\nabla \times \mathbf{F} = (-1, 0, -1)$$

$$\text{(d)} \quad \nabla \cdot \mathbf{F} = 1 + 3x + 1 = 3x + 2$$

$$\nabla \times \mathbf{F} = (0 - 0, 0 - 3y, 3y - 0) = (0, -3y, 3y)$$

9. Demonstrate the following vector identities if $f, g : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $\mathbf{F}, \mathbf{G} : \mathbb{R}^3 \rightarrow \mathbb{R}$.

(a) $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$

(b) $\nabla \cdot (f\mathbf{F}) = f\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla f$

(c) $\nabla \cdot (f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$

(d) $\nabla \cdot (\nabla f \times \nabla g) = 0$

$$\begin{aligned}
 \text{(a) } \nabla\left(\frac{f}{g}\right) &= \frac{\partial}{\partial x}\left(\frac{f}{g}\right) + \frac{\partial}{\partial y}\left(\frac{f}{g}\right) + \frac{\partial}{\partial z}\left(\frac{f}{g}\right) \\
 &= \frac{1}{g^2} \left(g \frac{\partial f}{\partial x} - f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial y} - f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial z} - f \frac{\partial g}{\partial z} \right) \\
 &= \frac{1}{g^2} \left(g \left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \right) - f \left(\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial g}{\partial z} \right) \right) \\
 &= \frac{g\nabla f - f\nabla g}{g^2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } \nabla \cdot (f\mathbf{F}) &= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (fF_1, fF_2, fF_3) \\
 &= \frac{\partial fF_1}{\partial x} + \frac{\partial fF_2}{\partial y} + \frac{\partial fF_3}{\partial z} \\
 &= f \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) \\
 &\quad + \left(F_1 \frac{\partial f}{\partial x} + F_2 \frac{\partial f}{\partial y} + F_3 \frac{\partial f}{\partial z} \right) \\
 &= f(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot \nabla f
 \end{aligned}$$

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$$(b) \nabla \cdot (f\mathbf{F}) = f\nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla f$$

$$(c) \nabla \cdot (f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$$

$$(d) \nabla \cdot (\nabla f \times \nabla g) = 0$$

$$(c) \nabla \cdot (f\nabla g - g\nabla f)$$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot [f \left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right) - g \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)]$$

$$= f \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right) - g \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right)$$

$$= f\nabla^2 g - g\nabla^2 f$$

$$(d) \nabla \cdot (\nabla f \times \nabla g)$$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot \left[\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) \times \left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right) \right]$$

$$= \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot \left(\frac{\partial f}{\partial y} \frac{\partial g}{\partial z} - \frac{\partial f}{\partial z} \frac{\partial g}{\partial y}, \frac{\partial f}{\partial z} \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} \frac{\partial g}{\partial z}, \frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} \right)$$

$$= \frac{\partial^2 f}{\partial x \partial y \partial z} + \frac{\partial^2 g}{\partial x \partial z \partial y} - \frac{\partial^2 f}{\partial x \partial z \partial y} - \frac{\partial^2 g}{\partial x \partial y \partial z} + \dots$$

$$= 0$$